

MICROBIOLOGY – SHORT EXAM NOTES

Sterilisation & Disinfection – Physical Methods, Chemical Methods, CSSD & Demonstration (Ananthanarayan & Paniker, Ch. 3)

1. INTRODUCTION AND DEFINITIONS

Term	Definition
Sterilisation	Process by which an article, surface or medium is freed of all living microorganisms, in vegetative or spore state
Disinfection	Destruction of all pathogenic organisms (not necessarily spores)
Antiseptics	Agents safely applied to skin or mucous membrane to prevent infection by inhibiting bacteria
Asepsis	State of complete absence of viable pathogenic microorganisms
Bactericidal / bacteriostatic	Kill bacteria / only prevent multiplication (a bactericidal chemical may become bacteriostatic at higher dilution)
Decontamination	Rendering an article or area free of contaminants – microbial, chemical, radioactive

Factors influencing sterilisation by heat: nature of heat (dry or moist), **temperature and time**, number of organisms, characteristics of organisms (species, strain, **spores**), and the material in which organisms are present (proteins, fats and oils increase thermal death time). **Thermal death time** is the minimum time required to kill a suspension of organisms at a given temperature.

2. PHYSICAL METHODS

Physical agents: **Heat** (dry and moist), **Filtration** and **Radiation**. Heat is the **most reliable** method. **Dry heat** kills by **oxidation** of cell constituents and protein denaturation; **moist heat** kills by **denaturation and coagulation of proteins** and is more efficient because steam releases **latent heat** on condensation and penetrates better.

A. Dry Heat

- **Flaming** – inoculating loops, wires, tips of forceps and searing spatulas held in a **Bunsen flame till red hot**; loops with infective material are dipped in disinfectant first to prevent spattering.
- **Incineration** – excellent method for destroying **biomedical waste** (soiled dressings, bedding, pathological material); PVC and polythene can be burnt, polystyrene gives toxic smoke.
- **Hot-air oven** – the most widely used dry-heat method; heat by **conduction**. **160°C for 2 hours** (170°C/1 h, 180°C/30 min). Used for **glassware, all-glass syringes, sharp instruments, liquid paraffin, dusting powder, fats and grease**. Fan for even air circulation; do not overload; glassware must be dry; **cool slowly (~2 h)** before opening.

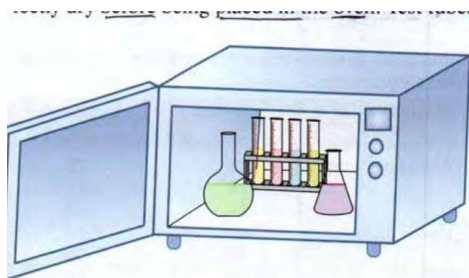


Fig. 3.1 Hot-air oven (from the book)

B. Moist Heat

Below 100°C

- **Pasteurisation of milk** – **holder method 63°C for 30 min** or **flash method 72°C for 15–20 s**, then rapid cooling to $\leq 13^{\circ}\text{C}$. Kills non-sporing pathogens (mycobacteria, salmonellae, brucellae); *Coxiella burnetii* may survive the holder method.
- **Inspissation** – heating at **80–85°C for ½ hour on 3 successive days** in an **inspissator**; for **Löwenstein–Jensen** and **Loeffler's serum** media.
- **Vaccine bath** – bacterial vaccines at **60°C for 1 hour**; **serum bath** – serum/body fluids at **56°C for 1 hour** on several successive days.

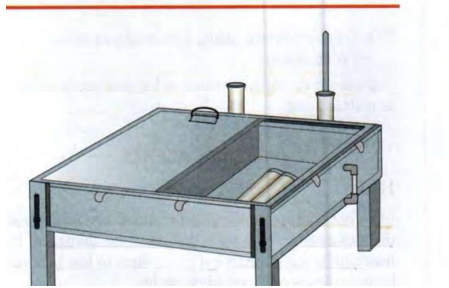


Fig. 3.2 Inspissator (from the book)

At 100°C

- **Boiling** – kills vegetative bacteria but **not spores**; boil 10–30 min; not recommended for surgical instruments; adding **2% sodium bicarbonate** softens hard water.
- **Steam at atmospheric pressure** – **Koch or Arnold steamer**; for media that decompose at higher temperature.
- **Tyndallisation (intermittent sterilisation)** – **100°C for 20 min on 3 successive days**; for media containing **sugars or gelatin**. First exposure kills vegetative forms; spores germinate and are killed on later exposures.

Above 100°C – Autoclave (Steam under Pressure)

Principle: water boils when its vapour pressure equals the surrounding pressure; raising pressure in a closed vessel raises the boiling point. Steam condensing on cooler articles releases **latent heat** (1600 ml steam $\rightarrow$ 1 ml water, 518 calories), and the volume reduction draws in more steam. Standard: **121°C, 15 lb/in², 15 minutes** (also 126°C/10 min, 134°C/3 min). Used for **culture media, linen, instruments, laboratory ware, gloves and infectious waste**.

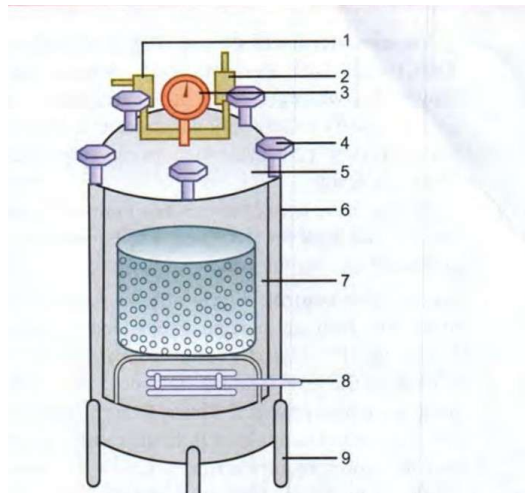


Fig. 3.3 A simple autoclave: 1. release valve 2. safety valve 3. pressure gauge 4. tightening screws 5. cover 6. body 7. bin containing material 8. heating element 9. stand (from the book)

Types: gravity displacement and **high-vacuum** sterilisers (air drawn out, faster penetration, post-cycle drying); laboratory autoclaves, dressing, instrument and rapid-cooling sterilisers. A **pressure cooker** is a miniature autoclave. **Defects of the simple autoclave:** inefficient air discharge and wet loads.

Recommended temperature and holding time (Table 3.1)

Method	Temperature	Holding time
Autoclave	121°C / 126°C / 134°C	15 / 10 / 3 min
Hot-air oven	160°C / 170°C / 180°C	120 / 60 / 30 min

C. Filtration

Removes bacteria from **heat-labile liquids** (sera, sugar and antibiotic solutions) and is used to obtain bacterial toxins. It makes fluids **bacteria-free but not virus-free**. Types: **candle filters** – unglazed ceramic (**Chamberland**, Doulton) and diatomaceous earth (**Berkefeld**, Mandler); **asbestos (Seitz)** – no longer used (carcinogenic); **sintered glass** – low absorption, brittle, expensive; **membrane filters** – cellulose esters, **0.22 µm** pore size, now most widely used for water analysis, sterility testing and parenteral solutions.

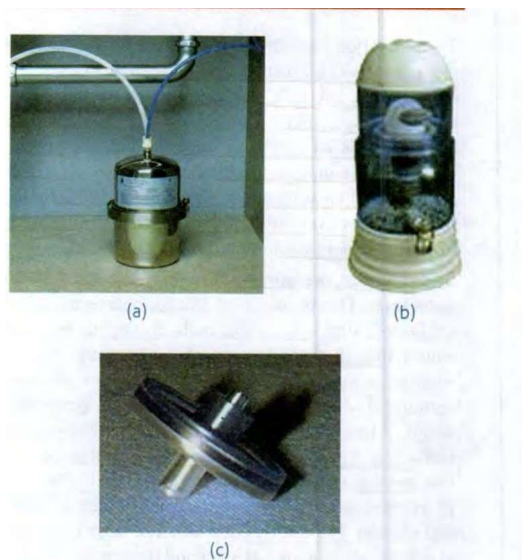


Fig. 3.4 (a) Asbestos filter (b) Sintered glass filter (c) Membrane filter (from the book)

D. Radiation

- **Non-ionising** (low energy): **infrared** – mass sterilisation of prepacked syringes and catheters; **ultraviolet** – disinfection of enclosed areas like **biosafety cabinets**, entryways, OTs and laboratories.
- **Ionising** (high energy, high penetration): **X-rays, gamma rays, cosmic rays** – lethal to DNA. No rise in temperature, hence called '**cold sterilisation**'. **Gamma rays** used commercially for plastics, syringes, swabs, catheters, animal feeds, oils, fabrics and metal foils.

Sterilisation Control

Method	Physical	Chemical	Biological indicator
Hot-air oven	Thermocouples	Browne's tube (green spot) – green after 60 min at 160°C	Spores of non-toxicogenic <i>C. tetani</i> or <i>B. subtilis</i> subsp. <i>niger</i>
Autoclave	Thermocouples, pressure gauge	Browne's tube, autoclave tape	Spores of <i>Geobacillus stearothermophilus</i> (killed at 121°C in 12 min)

3. CHEMICAL METHODS

Ideal antiseptic/disinfectant: wide spectrum; active in presence of organic matter; effective in acid and alkaline media; speedy action; high penetration; stable; compatible with other substances; non-corrosive, non-staining, non-toxic, and cheap. **Factors affecting potency:** concentration, contact time, pH, temperature, nature of organisms and presence of extraneous (organic) matter. **Modes of action:** protein coagulation, disruption of cell membrane, removal of sulphhydryl groups, and substrate competition for enzymes.

Important chemical agents

Agent	Action & features	Uses
Alcohols – ethyl, isopropyl (60–90%, best 70%); methyl	Denature proteins; no action on spores ; isopropyl better fat solvent	Skin antiseptic, clinical thermometers; methyl alcohol for cabinets and incubators
Formaldehyde	Acts on amino groups; bactericidal, sporicidal , virucidal; irritant gas	Preserving specimens; fumigation of wards/OTs (KMnO ₄ + formalin; not preferred now); neutralised by ammonia
Glutaraldehyde (2%)	Like formaldehyde, less toxic; active against TB bacilli, fungi, viruses; no damage to lenses	High-level disinfection of endoscopes, bronchoscopes, anaesthetic and polythene tubes
Orthophthalaldehyde, peracetic acid, H₂O₂, hypochlorous acid	Rapid bactericidal; H ₂ O ₂ fogging has replaced fumigation	Endoscopes; MRSA, VRE, <i>C. difficile</i> ; room fogging
Dyes – aniline (brilliant green, malachite green, crystal violet), acridine (acriflavine, proflavine)	Bacteriostatic; more active on Gram-positives; malachite green does not act on TB bacilli	Skin/wound antiseptics; selective agent in L-J medium
Halogens – iodine, iodophores (povidone-iodine 10%), chlorine, hypochlorite (0.5–5%)	Bactericidal, virucidal; iodine moderately sporicidal	Skin disinfection; water supply, pools; hospital disinfectant

Agent	Action & features	Uses
Phenols – phenol (5%), lysol, cresol, chlorhexidine (Hibitane), hexachlorophene	Membrane damage; not inactivated by organic matter (lysol/cresol)	Floors, walls, benches; chlorhexidine as skin antiseptic
Surface-active agents – cetrimide (Cetavlon), benzalkonium chloride	Cationic; act on cell membrane; inactivated by soap and acid; no action on spores, TB bacilli	Skin and wound antiseptics
Metallic salts – silver, copper, mercury (thiomersal, mercurochrome)	Protein coagulation, combine with SH groups	Mild antiseptics; copper as fungicide
Ethylene oxide (gas)	Alkylating agent ; highly penetrating, sporicidal; explosive (mixed with CO ₂ /N ₂), toxic, carcinogenic	Heat-sensitive equipment: heart-lung machines, respirators, sutures, dental equipment, plastics
Beta-propiolactone	Carcinogenic	No longer used

Testing of Disinfectants

- **Rideal–Walker test** – gives **phenol coefficient** (phenol = 1); ignores organic matter; now of historical interest.
- **Chick–Martin test** – modification in the presence of **organic matter** (dried yeast or feces).
- **Kelsey–Sykes / in-use test** – working solutions sampled during actual use in the hospital (practical test).

Spaulding's Classification

Category	Contact	Examples	Processing
Critical	Enter sterile tissue or vascular system	Surgical instruments, cardiac catheters, implants	Sterilisation – autoclave; ETO or H ₂ O ₂ plasma if heat-sensitive
Semi-critical	Mucous membranes or non-intact skin	Endoscopes, bronchoscopes	High-level disinfection (glutaraldehyde)
Non-critical	Intact skin only	Bedpans, BP cuffs, bed rails, tables	Cleaning / low-level disinfection

Levels of disinfectants: **High-level** – kills all pathogens except large numbers of spores (glutaraldehyde, H₂O₂). **Intermediate-level** – kills mycobacteria and non-enveloped viruses but not spores (alcohol, phenolics, iodophores). **Low-level** – kills only vegetative bacteria, fungi and enveloped viruses (quaternary ammonium compounds). **Plasma sterilisation** (hydrogen peroxide plasma, 'fourth state of matter') is used for heat-sensitive items like arthroscopes.

4. CENTRAL STERILE SUPPLY DEPARTMENT (CSSD)

The **CSSD** is the hospital department that **receives, cleans, packs, sterilises, stores and distributes** sterile instruments, linen and supplies to wards, OTs and other areas. Centralising the process ensures **uniform, quality-controlled sterilisation**, economy and reduced HAI. The **infection control nurse** of the HICC monitors CSSD practices.

Workflow (one-way, dirty → clean → sterile):

1. **Receiving** of used items in the dirty (decontamination) area.

2. **Cleaning and decontamination** – manual/ultrasonic washing with detergent (cleaning must precede sterilisation).
3. **Inspection, assembly and packing** – instrument sets wrapped with indicators.
4. **Sterilisation** – autoclave (main method), ETO or plasma for heat-sensitive items.
5. **Sterile storage** – clean, dry, dust-free area; first-in, first-out.
6. **Distribution** to user areas and record keeping.

Quality control: physical monitoring (time, temperature, pressure charts), chemical indicators (autoclave tape, Bowie–Dick test for air removal in vacuum autoclaves) and biological indicators (*G. stearothermophilus* spores).

The textbook only mentions CSSD under the Hospital Infection Control Committee; this section is standard practical content.

5. DEMONSTRATION (PRACTICAL SPOTTERS)

In the practical examination, identify the item and write its principle, use and control. The autoclave and hot-air oven procedures follow the book; the remaining steps are standard laboratory practice.

Spotter	Identify / principle	Key points to write
Autoclave	Moist heat – steam under pressure	121°C, 15 lb, 15 min; parts; expel air before closing tap; open only after pressure falls; control <i>G. stearothermophilus</i>
Hot-air oven	Dry heat – conduction, oxidation	160°C/2 h; glassware, oils, powders; cool slowly; Browne's tube, <i>B. subtilis</i> var. <i>niger</i>
Inspissator	Moist heat below 100°C	80–85°C, ½ h, 3 days; L–J medium, Loeffler's serum slope
Koch/Arnold steamer	Steam at 100°C	Tyndallisation – 100°C, 20 min, 3 days; sugar/gelatin media
Membrane / Seitz / candle filter	Filtration	Heat-labile fluids; 0.22 µm membrane; bacteria-free, not virus-free
Bunsen burner / spirit lamp	Dry heat – flaming	Loops, needles, forceps tips till red hot
UV lamp in BSC	Non-ionising radiation	Surface/air disinfection of cabinets and rooms
Browne's tube / autoclave tape	Chemical indicators	Colour change confirms exposure to sterilising conditions
Spore strip (biological indicator)	Biological control	<i>G. stearothermophilus</i> (autoclave), <i>B. subtilis</i> var. <i>niger</i> (HAO); no growth after incubation = sterile
Disinfectants (70% alcohol, 2% glutaraldehyde, 1% hypochlorite, 5% phenol, povidone-iodine)	Chemical methods	Name, concentration, mechanism and uses

Demonstration Steps – Autoclave

1. Check water level in the chamber; load material in the bin (not overcrowded; plugs loosened, media not over-filled).
2. Close the lid, tighten screws evenly, keep the **discharge tap open** and switch on heating.
3. Allow steam–air mixture to escape until **all air is expelled** (no more bubbles), then close the tap.
4. When pressure reaches **15 lb/in² (121°C)** and the safety valve blows, start **holding time of 15 min**.

5. Switch off; let pressure fall to **atmospheric (zero)**; open the discharge tap slowly, then the lid.
6. Check the chemical indicator; record; periodically use spore strips.

Demonstration Steps – Hot-Air Oven

1. Load **clean, dry**, wrapped glassware with space for air circulation.
2. Switch on; set thermostat to **160°C**; start timing when 160°C is reached – hold **2 hours**.
3. Switch off and allow to cool **slowly (~2 h)** before opening; check Browne's tube.

★ EXAM FOCUS

- Essays: **Define sterilisation; principle and working of autoclave; Dry heat methods & hot-air oven; Disinfection techniques in hospitals.**
- Short notes: **Pasteurisation, Tyndallisation, Inspissator, Filtration, Gaseous sterilisation (ETO), Cold sterilisation, Sterilisation control, Spaulding's classification, Plasma sterilisation, Testing of disinfectants.**